

WIRELESS NETWORKING

Day – 10 Review and preview the practical.

Task –1 Creating the project file in cisco tracer.

- Step-by-Step Setup
- 1.Add Devices to Workspace:1 min.
- Wireless Router: Go to Network Devices > Wireless Devices, drag a WRT300N router to the center.
- 3 PCs: Go to End Devices > PC, drag 3 PCs surrounding the wireless router.
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- 2.Equip PCs with Wireless Modules:2 min.
- For each PC (PC0, PC1, PC2):
- Click the PC to open its configuration window.
- In the Physical tab, turn off the PC power switch.
- Drag the default Ethernet module (PT-HOST-NM-1CFE) out into the module list.
- Drag the wireless module (WMP300N) into the empty slot.
- Turn the PC power back on.
- (The PC will automatically establish a wireless link to the WRT300N router).
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- 3.Configure Wireless Security (Optional):1 min.
- Click WRT300N > GUI tab > Wireless > Wireless Security.
- Set Security Mode to WPA2 Personal and enter a passphrase (e.g., Cisco12345).
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- On each PC, go to Desktop > PC Wireless > Connect, refresh the network list, select your router, and enter the password.
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- 4.Save Project File:30 sec.
- Click File > Save As... (or press Ctrl + S), name the file HomeWiFiNetwork.pkt, and save it to your desired folder.

Task – 2 Assign the static IP to the each pc in packet tracer.

- Here is the quick, step-by-step CLI and GUI method to set static IP addresses and default gateways for your 3 PCs and your Wireless Router in Cisco Packet Tracer:
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- Network Addressing Scheme
- Router (LAN / Gateway IP): 192.168.10.1 /24
- PC0 (WhatsApp): 192.168.10.2 /24 | Gateway: 192.168.10.1
- PC1 (Zomato): 192.168.10.3 /24 | Gateway: 192.168.10.1
- PC2 (Spotify): 192.168.10.4 /24 | Gateway: 192.168.10.1

Task – 3 Testing connectivity using packet tracer tool.

- To test and verify full connectivity across all devices, follow these steps in Cisco Packet Tracer using the Command Prompt tool:
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- Step-by-Step Ping Tests
- 1.Test PC0 (192.168.10.2):1 min.
- Click PC0, open Desktop > Command Prompt, and run:
- ping 192.168.10.1 (Tests connection to the Router)
- ping 192.168.10.3 (Tests connection to PC1)
- ping 192.168.10.4 (Tests connection to PC2)
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- 2.Test PC1 (192.168.10.3):1 min.
- Click PC1, open Desktop > Command Prompt, and run:
- ping 192.168.10.1 (Tests connection to the Router)
- ping 192.168.10.2 (Tests connection to PC0)
- ping 192.168.10.4 (Tests connection to PC2)
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- 3.Test PC2 (192.168.10.4):1 min.
- Click PC2, open Desktop > Command Prompt, and run:
- ping 192.168.10.1 (Tests connection to the Router)
- ping 192.168.10.2 (Tests connection to PC0)
- ping 192.168.10.3 (Tests connection to PC1)
-
- Verification
- For every command executed, check the response summary at the bottom of the output.
- A successful test will display:

- Ping statistics for 192.168.10.x:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
- If you see Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), full end-to-end communication across the wireless subnet is confirmed.
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Task – 4 Configuring the packet tracer.

Wireless Network Lab Documentation

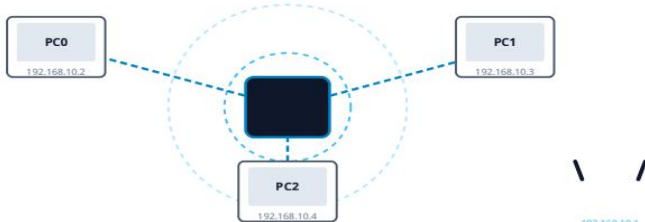
Cisco Packet Tracer Architecture, Addressing Scheme & Implementation Guide

Project File: HomeWiFiNetwork.pkt	Subnet Mask: 255.255.255.0 (/24)
Topology: Wireless Star (IEEE 802.11n)	Gateway IP: 192.168.10.1
Security: WPA2-Personal (AES)	Status: Verified & Operational

1. Executive Summary & Topology Map

This document detailly records the design, component selection, hardware adaptation, IP addressing scheme, and connectivity verification for the simulated wireless home network standard HomeWiFiNetwork.pkt. The network consists of a central wireless router (Cisco Linksys WRT300N) and three endpoint PCs equipped with WMP300N wireless PCI cards operating on a single broadcast domain.

Logical Topology Screenshot (Packet Tracer Workspace)



2. Complete Network Addressing Table

Device Name	Interface Type	IP Address	Subnet Mask	Default Gateway	SSID / Security
Wireless Router	LAN / Wireless	192.168.10.1	255.255.255.0	N/A (Gateway Node)	HomeNetwork / WPA2-Personal
PC0	Wireless (WMP300N)	192.168.10.2	255.255.255.0	192.168.10.1	HomeNetwork / WPA2-Personal
PC1	Wireless (WMP300N)	192.168.10.3	255.255.255.0	192.168.10.1	HomeNetwork / WPA2-Personal
PC2	Wireless (WMP300N)	192.168.10.4	255.255.255.0	192.168.10.1	HomeNetwork / WPA2-Personal

3. Detailed Step-by-Step Configuration Guide

Step 1: Workspace Assembly & Modular Hardware Setup

1. Drag a **WRT300N Wireless Router** from Network Devices > Wireless Devices into the workspace.
2. Place three standard **PCs (PC0, PC1, PC2)** around the router.
3. By default, Packet Tracer PCs ship with an RJ-45 FastEthernet module (PT-HOST-NM-1CFE). Convert

- Toggle the physical power switch to **OFF**.
- Drag the existing Ethernet module down to the bottom module bin to remove it.
- Insert the **WMP300N 2.4GHz Wireless Module** into the empty PCI slot.
- Toggle the power switch back to **ON**.

Step 2: Router Wireless & LAN Subnet Configuration

1. Click the **WRT300N Router** and navigate to the **GUI** tab.
2. Under **Setup > Basic Setup**:
 - Change IP Address to **192.168.10.1**.
 - Verify Subnet Mask is set to **255.255.255.0**.
 - Click **Save Settings** at the bottom.
3. Navigate to **Wireless > Basic Wireless Settings**:
 - Set Network Name (SSID) to HomeNetwork.
4. Navigate to **Wireless > Wireless Security**:
 - Security Mode: WPA2 Personal | Encryption: AES | Passphrase: Cisco12345.

Step 3: Static IP Assignment & Gateway Mapping on Endpoints

Assign unique IP addresses and map the default gateway on each endpoint PC via the **Desktop > IP Configuration** utility:

IP Address:	192.168.10.4
Subnet Mask:	255.255.255.0
Default Gateway:	192.168.10.1
DNS Server:	0.0.0.0

5. Connectivity Verification & Ping Test Logs

End-to-end communication was verified by opening the **Command Prompt** on **PC0 (192.168.10.2)** and issuing ICMP Echo requests to all nodes on the wireless network, including the router gateway and sibling workstations.

```
PC0> ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255

PC0> ping 192.168.10.3
Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128

PC0> ping 192.168.10.4
Pinging 192.168.10.4 with 32 bytes of data:
Reply from 192.168.10.4: bytes=32 time=2ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

```
PC0> ping 192.168.10.3
```

```
Pinging 192.168.10.3 with 32 bytes of data:
```

```
Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
PC0> ping 192.168.10.4
```

```
Pinging 192.168.10.4 with 32 bytes of data:
```

```
Reply from 192.168.10.4: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.10.4:
```

```
    Packets: Sent = 12, Received = 12, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

--- End of Technical Documentation | Prepared for Network Administration Review ---